

$$\frac{du}{dt} + Au = 0$$

$$u = e^{\lambda t} \vec{v}$$

$$|\lambda I + A| = 0$$

$$\frac{dx}{dt} + \frac{3}{2}x + \frac{5}{2}y = 0$$

$$\frac{dy}{dt} - \frac{5}{2}x - \frac{3}{2}y = 0$$

$$\begin{vmatrix} \frac{3}{2} + \lambda & \frac{5}{2} \\ -\frac{5}{2} & -\frac{3}{2} + \lambda \end{vmatrix} = \left(\frac{3}{2} + \lambda\right)\left(-\frac{3}{2} + \lambda\right) + \frac{25}{4} = 0$$

$$\left(\lambda^2 - \frac{9}{4}\right) + \frac{25}{4} = 0$$

$$\lambda^2 = -\frac{25}{4} + \frac{9}{4} = -\frac{16}{4} = -4$$

$$\underline{\lambda = \pm 2i}$$

$$\Rightarrow u = \begin{pmatrix} a \\ b \end{pmatrix} e^{2it}$$

$$\begin{pmatrix} \frac{3}{2} + 2i & \frac{5}{2} \\ -\frac{5}{2} & -\frac{3}{2} + 2i \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0$$

$$-\frac{5}{2}a + \left(-\frac{3}{2} + 2i\right)b = 0$$

$$-5a + (4i - 3)b = 0$$

$$a = \frac{(4i - 3)b}{5}; \quad b = 5$$

$$u_1 = \begin{pmatrix} 4i - 3 \\ 5 \end{pmatrix} e^{2i}; \quad u_2 = \begin{pmatrix} 4i + 3 \\ 5 \end{pmatrix} e^{-2i}$$

$$u = \begin{pmatrix} -3 \\ 5 \end{pmatrix} e^{2i} + i \begin{pmatrix} 4 \\ 0 \end{pmatrix} e^{2i}$$

$$u = \begin{pmatrix} -3 \\ 5 \end{pmatrix} [\cos 2t + i \sin 2t] + i \begin{pmatrix} 4 \\ 0 \end{pmatrix} [\cos 2t + i \sin 2t]$$

$$u = \begin{pmatrix} -\frac{3}{5} \\ 1 \end{pmatrix} \cos 2t - \begin{pmatrix} 4 \\ 0 \end{pmatrix} \sin 2t + i \left\{ \begin{pmatrix} -3 \\ 5 \end{pmatrix} \sin 2t + \begin{pmatrix} 4 \\ 0 \end{pmatrix} \cos 2t \right\}$$

$$u = C_1 \left[\begin{pmatrix} -3 \\ 5 \end{pmatrix} \cos 2t - \begin{pmatrix} 4 \\ 0 \end{pmatrix} \sin 2t \right] + \left\{ \begin{pmatrix} -3 \\ 5 \end{pmatrix} \sin 2t + \begin{pmatrix} 4 \\ 0 \end{pmatrix} \cos 2t \right\}$$

$$\frac{d^2 x}{dt^2} - \frac{dy}{dt} - x + 3y = 0$$

• Ecuación inhomogénea.

$$\frac{dx}{dt} + \frac{dy}{dt} + x - y = 0$$

• Ecuaciones de hamilton

Podríamos proponer $\frac{dx}{dt} = z$

Tenemos...

$$\frac{dy}{dt} + z + x - y = 0$$

$$\frac{dz}{dt} + z + \cancel{-y} - x + 3y = 0$$

$$\frac{dz}{dt} + z + 2y = 0$$

$$\rightarrow \frac{dx}{dt} - z = 0$$

$$\frac{dy}{dt} + z + x - y = 0$$

$$\frac{dz}{dt} + z + 2y = 0$$

$$u = \begin{pmatrix} x \\ y \\ z \end{pmatrix} ; \quad A = \begin{pmatrix} 0 & 0 & -1 \\ 1 & -1 & 1 \\ 0 & 2 & 1 \end{pmatrix}$$

Tenemos

$$\frac{dx}{dt} - 2x - 6y = 2t - 2 - e^{-t}$$

$$\frac{dy}{dt} + 2x + 5y = 2e^t - 3t + 3$$

$$u = c_1 \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} -3 \\ 2 \end{pmatrix} e^{-2t}$$

$$u = \gamma(t) \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-t} + \gamma_2(t) \begin{pmatrix} -3 \\ 2 \end{pmatrix} e^{-2t}$$

Tenemos

$$\frac{du}{dt} + Au = \begin{pmatrix} 2t - 2 - e^{-t} \\ 2e^t - 3t + 3 \end{pmatrix}$$

$$x_1' \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-t} - \begin{pmatrix} -2 \\ 1 \end{pmatrix} x_1 e^{-t} + x_2' \begin{pmatrix} -3 \\ 2 \end{pmatrix} e^{-2t} - 2 x_2 \begin{pmatrix} -3 \\ 2 \end{pmatrix} e^{-2t}$$

$$\Rightarrow x_1 \begin{pmatrix} -2 & -6 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-t} + x_2 \begin{pmatrix} -2 & 6 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} -3 \\ 2 \end{pmatrix} e^{-2t} =$$

• Caso para equações
de Hamilton.

$$\begin{pmatrix} 2t - 2 - e^{-t} \\ 2e^t - 3t + 3 \end{pmatrix}$$

$$\Leftrightarrow -2x_1' e^{-t} - 3x_2' e^{-2t} = 2t - 2(-e^{-t})$$

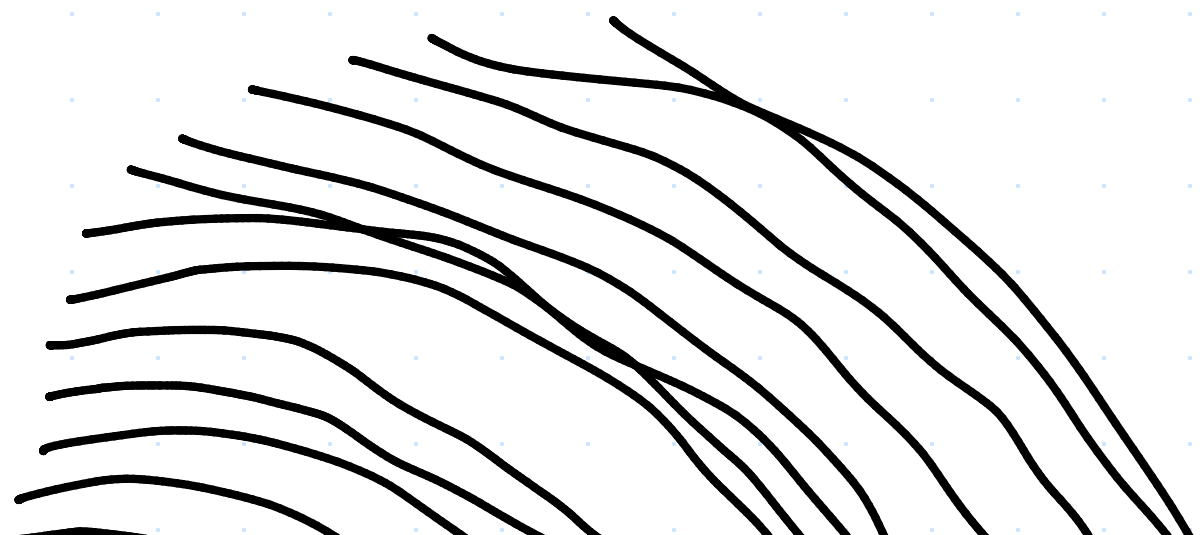
$$(x_1' e^{-t} + 2x_2' e^{-2t} = 2e^t - 3t + 3) \cdot 2$$

Temos ...

$$x_2' e^{-2t} = -4t + 4 - e^{-t} + 4e^t$$

- Teoria quântica de campos
- corpo rígido

} Temas optativas



• Coordenadas curvilíneas → Examen.

→ Martes 2 de septiembre examen.

Ecuaciones parciales

$$\frac{\partial^2 u}{\partial x \partial y} = 6x + 12y^2 ; \quad u(1, y) = y^2 - 2y ; \quad u(x, 2) = 5x - 5$$

$$\frac{\partial u}{\partial y} = 3x^2 + 12y^2 x + f(y)$$

$$u(x, y) = 3x^2 y + 4xy^3 + \underbrace{\int f(y) dy}_{F(y)} + H(x)$$

Tenemos

$$u(1, y) = y^2 - 2y = 3y + 4y^3 + F(y) + H(1)$$

$$F(y) = y^2 - 4y^3 - 5y - H(1)$$

$$u(x, y) = 3x^2 y + 4xy^3 + y^2 - 4y^3 - 5y + H(1) + H(x)$$

$$5x - 5 = 6x^2 + 32x + 4 - 32 - 10 + H(1) + H(x)$$

$$5x - 32x - 5 + 38 - 6x^2 - H(1) = H(x)$$

$$H(x) = -27x + 33 - 6x^2 - H(1)$$

$$u(x, y) = 3x^2y + 4xy^3 + y^2 - 4y^3 - 5(y) + H(1) - 27x + 33 - 6x^2 - H(1)$$

Otro ejemplo

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial u}{\partial x} + 2 ; \quad u(0, y) = 0 ; \quad u'(x, y) \Big|_{y=0} = x^2$$

$$\frac{\partial}{\partial x} \left[\frac{\partial u}{\partial y} - u \right] = 2$$

$$\Rightarrow \frac{\partial u}{\partial y} - u = 2x + F(y)$$

Tenemos viendolo como producto

$$\underbrace{e^{-y} \frac{\partial u}{\partial y} - u e^{-y}} = 2x e^{-y} + F(y) e^{-y}$$

$$\frac{\partial}{\partial y} (u e^{-y})$$

$$\rightarrow u e^{-y} = -2x e^{-y} + \underbrace{\int f(y) e^{-y} dy}_{F(y)} + H(x)$$

$$u = -2x + F(y) + H(x) e^y$$

Tenemos

usando las condiciones iniciales

$$u(0, y) = 0$$

$$0 = F(y) + H(0) e^y ; \quad F(y) = -e^y H(0)$$

$$u = -2x + H(x) e^y - e^y H(0)$$

$$\left. \frac{d}{dx} u \right|_{y=0} = -2 + H'(x) e^y \Big|_{y=0}$$

$$H'(x) = 2 + x^2$$

$$H(x) = 2x + C + \frac{x^3}{3}$$

$$u = -2x + 2xe^y + Ce^y - e^y H(0) + \frac{x^3}{3}$$

$$\underline{C = H(0)}$$

Divergencia en variable compleja